Abhilash Balachandran

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Professional Summary

AI Engineer with 8+ years of software and ML experience, specializing in production-grade GenAI systems: multi-agent architectures, RAG, eval pipelines, and enterprise-scale deployment. Shipped production AI systems driving $2M+ in annual recurring revenue.

Work Experience

Amazon Web Services Inc. **Mountain View, CA**
Senior AI Engineer | Deep Learning Architect Sept 2024 - Present

- Enterprise Conversational AI Platform
  - Architected and built a reusable multi-channel conversational agent framework for Fortune 500 customers, automating complex workflows across voice, web, SMS, and WhatsApp; powers 5+ enterprise deployments totaling $2M+ ARR
  - Built the multi-agent backend shared across all channels, designing prompts, tool interfaces, and orchestration for specialized agents: a RAG agent grounded in enterprise knowledge sources and other reasoning and tool-calling agents for workflow automation
  - Designed a dual-agent streaming architecture for real-time channels: the multi-agent backend runs complex reasoning and tool calls in the background while a foreground agent streams contextual updates to the user, eliminating latency dead zones between responses
  - Built a CI/CD evaluation pipeline using LLM-as-User to generate simulated multi-turn conversations at scale and LLM-as-Judge to score them, validating agent configuration and prompt changes before production release
  - Engineered the platform for fault tolerance and production observability: queue-backed async workers with retries, per-agent checkpointing, async online evaluation, distributed tracing, and operational dashboards
- Multi-Agent Game Generation System
  - Built multi-agent game generation harness - users describe a game concept and specialized agents collaboratively generate themed assets, graphics, and content
  - Agentic harness reduced content creation time from 8 hours to 1 hour with 60% cost reduction
- Developer AI Tooling
  - Applied AIDLC (AI-Driven Development Life Cycle) patterns for customer delivery, using spec-driven development with coding assistants against production architecture standards to accelerate delivery 4x across projects
  - Owned end-to-end infrastructure for a production codebase Q&A system on ECS (Cognito auth, WebSocket streaming, CDK), then extended it as an MCP server for direct integration into AI coding environments
  - Built agentic skills for automating developer workflows: security fix generation, metrics instrumentation, cost tracking, and observability dashboarding
  - Contributed to fine-tuning open-source and proprietary LLMs using SFT with LoRA for structured content generation

Ridecell Inc. **Milpitas, CA**

Senior Software Engineer, AI Oct 2022 - Sept 2024

- Built LLM-based customer support system using RAG **-** demonstrated 12x reduction in troubleshooting time; deployed on EKS with Weaviate vector DB for hybrid search
- Built and benchmarked Text-to-SQL pipeline for natural language querying of vehicle telemetry data - evaluated GPT-powered vs proprietary model solutions
- Fine tuned LLMs for Text-to-SQL code generation
- Validated and deployed AI-powered vehicle damage assessment system - damage type, part, severity, and repair cost estimation
- Led the development of MLOps infrastructure for automated model training and deployment using Airflow and Kubeflow

Software Engineer, Data & ML Oct 2020 - Sept 2022

- Built driver behavior scoring model from GPS, acceleration, and road data - processed 20GB+ vehicle data and enabled dynamic pricing that generated $100K in annual savings

Software Engineer, Autonomous Driving May 2018 - Sept 2020

- Developed Computer Vision modules leveraging deep learning and ML for autonomous driving including real-time pedestrian and object detection

Open Source Contributions & Publications

- Contributor: Fullstack AgentCore Solution Template - Open-sourced production-ready template for deploying full-stack agentic applications on AgentCore
- Author: indexed-repository-intelligence-system - Open-sourced agentic intelligence system built on Amazon Bedrock AgentCore that reasons over codebases and documents using a suite of tools to deliver grounded, citation-backed answers to natural-language questions
- Featured Author, AWS Blog: AI-powered virtual agents: Automating complex business processes

Education

Johns Hopkins University **Baltimore, MD**

Master of Science in Engineering – Robotics Sept 2016 - Dec 2017

National Institute of Technology **Tiruchirappalli, India**

Bachelor of Technology – Mechanical Engineering July 2011 - Aug 2015

Technical skills

- **Languages:** Python, SQL
- **Backend:** FastAPI, Flask, REST APIs, WebSocket Streaming
- **Infrastructure:** Docker, CI/CD, ECS, EKS, Kubernetes, Lambda, CDK, Cognito, CloudFront
- **Data & MLOps:** PySpark, Pandas, MLflow, Airflow, Kubeflow, SageMaker
- **AWS Services:** Bedrock, DynamoDB, S3, Connect, SQS, SNS
- **ML & Modeling:** PyTorch, SFT, LoRA, Fine-tuning
- **Generative AI & Agents:** LangGraph, LangChain, MCP, RAG, Vector DBs (Chroma, OpenSearch, Pinecone, Weaviate), DSPy
- **AI Dev Tools:** Claude Code, Codex, Kiro
- **Skills:** Agentic Architecture Patterns, Agentic Evaluation, Multi Agent Systems, Distributed Systems, Machine Learning